# Unit V
# 13. Renewable energy sources

Renewable energy refers to energy derived from natural resources that are continuously replenished on a human timescale — such as sunlight, wind, water, and geothermal heat.

Using renewable energy instead of fossil fuels helps organizations:
- Minimize greenhouse gas emissions,
- Reduce dependence on non-renewable sources, and
- Promote environmental sustainability

**13.1 Solar energy**
Solar energy is harnessed from the sun's radiation using technologies such as photovoltaic (PV) panels and solar thermal systems.

**Working Principle:**
- **Solar Photovoltaic (PV) Systems:** Convert sunlight directly into electricity using semiconductor materials (like silicon).
- **Solar Thermal Systems:** Use mirrors or collectors to concentrate sunlight and produce steam for power generation.

**Advantages:**
- Abundant and freely available source.
- Zero emissions during operation.
- Low maintenance costs.

**Challenges:**
- Intermittent power generation (dependent on sunlight).
- High initial installation cost.
- Large area required for solar farms.

**13.2 Wind Energy**
Wind energy is generated by converting the kinetic energy of moving air into electrical energy using wind turbines.

**Working Principle:**
- Wind turns the blades of a turbine connected to a generator.
- The mechanical rotation produces electrical power through electromagnetic induction.

**Advantages:**
- Clean and renewable.
- Suitable for large-scale power generation.
- Reduces dependence on fossil fuels.

**Challenges:**
- Wind variability and location constraints.
- Noise and visual impact.
- Requires significant land or offshore installations.

**13.3 Hydro Energy**

Hydropower or hydroelectric energy uses the potential energy of stored water (usually in dams) to drive turbines and generate electricity.

**Working Principle:**
- Flowing or falling water rotates a turbine connected to a generator.
- The mechanical energy is converted into electrical energy.

**Advantages:**
- Reliable and consistent energy source.
- High efficiency (up to 90%).
- Provides grid stability and energy storage through pumped hydro systems.

**Challenges:**
- Environmental impact due to dam construction.
- Displacement of local communities and ecosystems.
- Limited to suitable geographical locations.

**13.4 Geothermal Electricity Generation**
Geothermal energy is derived from the heat stored beneath the Earth's surface, harnessed for generating electricity or direct heating.

**Working Principle:**
- Wells are drilled into geothermal reservoirs to bring hot water or steam to the surface.
- The steam drives turbines connected to generators to produce electricity.

**Advantages:**
- Reliable and available 24/7 (not weather-dependent).
- Minimal land requirement.
- Low emissions compared to fossil fuels.

**Challenges:**
- Limited to regions with geothermal activity.
- High upfront drilling and installation costs.
- Potential release of underground gases.

# 14. Case study

Successful e-waste management and recycling initiatives in the tech industry.

**Case 1: Dell Technologies – "Closed-Loop Recycling System"**

Dell is recognized as one of the pioneers in circular electronics design and e-waste recycling.
Its "Closed-Loop Recycling" initiative focuses on reclaiming materials from used electronics and reusing them in new products.

**Key Initiatives**

- Dell Reconnect Program: A partnership with Goodwill Industries that allows consumers to drop off old computers and accessories for free recycling.
- Closed-Loop Supply Chain: Plastics recovered from recycled electronics are reintroduced into the production of new Dell laptops, desktops, and monitors.
- Responsible Recycling: Uses only R2 (Responsible Recycling) and E-Stewards certified recyclers.

**Impact**

- Recycled over 2.5 billion pounds of electronics since 2007.
- Reduced use of virgin plastics by reusing recovered materials.
- Lowered carbon footprint across the supply chain.

# 15. Latest Green Technologies and Emerging Trends

Green computing means using computers and technology in a way that's better for the environment. New trends include energy-saving devices, using renewable energy for datacenters, and recycling old hardware. AI helps make systems more efficient by reducing energy use. Companies are also focusing on reusing and repairing tech to cut down on waste.

For example, Google runs its data centers with 100% renewable energy. Another trend is using AI to manage energy better, like adjusting server use and cooling systems to save power.
Edge computing helps by processing data closer to where it's needed, which saves energy — for example, Tesla's Autopilot processes data in the car instead of sending it to the cloud.
Companies like Apple and Dell are also working on a circular economy by recycling old devices and using recycled materials in new products. Even blockchain is becoming greener — for example, Ethereum switched to a new method called Proof of Stake that uses 99% less energy than the old system.

## 15.1 Overview of the latest trends in green computing

- **Cloud computing and its environmental implications**

Cloud computing is the delivery of computing services like storage, processing, and software over the internet instead of using a personal computer or local server. Instead of storing files or running programs on your own computer, you access them online through services like Google Drive, Dropbox, or Microsoft Azure. The "Cloud" refers to large data centers with powerful computers that store and process this information, allowing you to access it from anywhere with an internet
connection.

**How does switching to hyperscale datacenters help reduce the environmental impact of cloud computing?**
Switching to hyperscale datacenters helps reduce the environmental impact because these facilities are designed to be more energy-efficient and use renewable energy sources like solar and wind. Hyperscale datacenters can handle large amounts of data with fewer servers, reducing overall electricity consumption. They also use advanced cooling systems and AI to manage energy use more efficiently. According to studies, shifting to hyperscale facilities could cut cloud energy use by up to 25%, lowering carbon emissions and reducing the environmental footprint.

**What is hyperscale datacenter?**
A hyperscale datacenter is a large facility designed to handle huge amounts of data and computing power efficiently. It uses advanced technology to save energy, like smart cooling systems and AI-based management. Big companies like Google, Amazon, and Microsoft use hyperscale datacenters to power their cloud services.

- **Edge computing and IoT for sustainability**

- **Edge + IoT** shift some processing nearer data sources to reduce network transfer and improve latency — when designed smartly, this can lower energy use and network carbon footprint (e.g., local aggregation, event filtering, intermittent sync).

- IoT sensor networks enable real-time environmental monitoring (air, water, grid), allowing targeted interventions (smart grids, precision agriculture) that reduce overall resource use. (See circular / smart-city pilots and waste-reduction projects.)

❑ **Blockchain technology for green initiatives**

- Blockchain is like a digital notebook where we write down important information that everyone can see, but nobody can change, like a magical diary.
- Example: Imagine a big book where everyone writes their names when they borrow book from a library, and nobody can erase or change what they wrote.
- Unlike a traditional ledger that is controlled by one person or company, a blockchain is decentralized, meaning everyone in the network has a copy of the same data. This makes it more secure and trustworthy because if someone tries to change the data in one place, the other computers in the network will notice the difference and reject the change. This decentralized nature helps prevent fraud and hacking, as the data must match across all copies of the ledger for any change to be accepted.
- Blockchain is used for traceability and verification in renewable certificates, carbon credits, and circular supply chains — useful for provenance of recycled materials and verifying renewable energy procurement. However, energy costs of some consensus mechanisms remain a concern, so proof-of-stake and lighter ledgers are preferred for green use cases.

❑ **Artificial intelligence for energy optimization**

- **Artificial Intelligence (AI)** is the ability of machines and computers to think, learn, and make decisions like humans. AI uses algorithms and data to analyze patterns, solve problems, and improve over time without human input. It powers technologies like voice assistants (e.g., Siri), self-driving cars, and facial recognition.
- AI is being deployed to optimize datacenter operations (cooling, workload placement), grids (demand forecasting), and building energy management. Google/DeepMind's ML for cooling is a canonical example (large % cooling reduction). Expect more AI tools that balance performance vs. energy in real time.

❑ **Circular economy and sustainable IT practices.**

- The circular economy is designed for repair, reuse, remanufacture, and material recovery and hence is moving into IT: closed-loop plastics, take-back programs, modular devices, and material recovery at scale. Policy and industry programs are accelerating adoption.

# 16. Case Study

Future directions and challenges in green computing, including research opportunities and industry trends

**1) Google: engineering the 24/7 carbon-free datacenter**
Google set a goal to operate on carbon-free energy 24/7 in every region by 2030 and is investing in long-duration storage, direct partnerships (Energy Dome), and even ordered small modular reactors to secure reliable low-carbon power. They also report operational emissions improvements (e.g., a reported 12% reduction in data-center energy emissions even as consumption grew). These moves shift focus from buying annual renewable credits to matching energy on an hourly basis and securing firm low-carbon generation.
**Why it matters:** 24/7 carbon-free operation is a systems problem — it requires new methods for procurement, long-duration storage, and sometimes novel generation (SMRs), and it directly affects how cloud customers (and research projects) get "clean" compute.

**2) Dell : circular design and large-scale e-waste recovery**
Dell has invested in circular design (modular laptops, reclaimed plastics), large reverse-logistics recycling programs, and closed-loop supply chains; since 2007 the company reports recovering billions of pounds of electronics for reuse or material recovery.
**Why it matters:** Extending product life, designing for repair, and recovering critical materials reduce demand for virgin materials and lower lifecycle emissions — but they require new business, regulatory and technical practices (modularity, standardized repairability, tracking returned materials).